Supplemental Material: The Cavity-Loss Barrier to Ultrastrong-Coupling Quantum Metrology

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This Supplemental Material provides the model definitions and QFI conventions (Section S1), numerical convergence (Section S2), the single-qubit and collective loss calculations that produce the main retention values (Sections S3 and S4), the cavity-channel HEOM cross-check (Section S5), the multilevel-emitter leakage screen (Section S6), the finite-size scaling (Section S7), and the additional model-specific screens and feasibility considerations that support the main-text boundaries (Sections S8 and S9).

S1. MODEL DEFINITIONS AND QFI CONVENTIONS

We set $\hbar = 1$ throughout. The single-qubit quantum Rabi Hamiltonian is

$$\hat{H}_{\mathrm{Rabi}} = \frac{\omega_q}{2}\hat{\sigma}_z + \omega_c \hat{a}^\dagger \hat{a} + g\hat{\sigma}_x(\hat{a} + \hat{a}^\dagger), \tag{S1}$$

with normalized coupling $\eta = g/\omega_c$. The collective Dicke Hamiltonian for N qubits is

$$\hat{H}_{\mathrm{Dicke}} = \omega_c \hat{a}^\dagger \hat{a} + \omega_q \hat{J}_z + \frac{g}{\sqrt{N}}\hat{J}_x(\hat{a} + \hat{a}^\dagger), \tag{S2}$$

with $\hat{J}_{x,z} = \frac{1}{2}\sum_{i=1}^{N} \hat{\sigma}_{x,z}^{(i)}$. The $1/\sqrt{N}$ normalization gives the thermodynamic critical coupling $\eta_c = \frac{1}{2}\sqrt{\omega_q/\omega_c}$, which equals 0.5 at resonance.

Both Hamiltonians are diagonalized in the Fock-qubit (or collective-spin) basis with truncations $n_{\max} = 50$ (Rabi) and $n_{\max} = 35$ (Dicke, $N \leq 64$). The driven DCE protocol modulates the coupling as $g(t) = g_0 + g_d \cos(\omega_d t)$; the Schrödinger equation is integrated with QuTiP [1]. Because the interaction is time dependent, strongly nonadiabatic USC control is gauge relative [2], and the results below are interpreted as effective-control-model predictions, not gauge-independent device forecasts. A controlled finite-dimensional similarity transformation of Hamiltonian, state, control, and observables (see Section S8) verifies internal representation consistency to 5×10^{-9} but does not resolve the physical gauge ambiguity.

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A. Quantum Fisher information

For a static eigenstate family $\hat{H}(\eta)\,|\Psi_G(\eta)\rangle = E_0(\eta)\,|\Psi_G(\eta)\rangle$ with generator $\hat{H}_{\rm int} = \partial_\eta \hat{H}$, the exact QFI is the gap-weighted sum

$$F_Q = 4\sum_{k\neq 0} \frac{|\,\langle\Psi_k|\hat{H}_{\rm int}|\Psi_G\rangle\,|^2}{(E_k - E_0)^2}. \tag{S3}$$

The variance form $F_Q = 4\,\mathrm{Var}[\hat{H}_{\rm int}]$ holds only for unitary-encoding families and overestimates the static QFI by a factor 3.4 at the sampled maximum (20.9 vs. 6.116 at $\eta = 1.14$). The reported static values are the fidelity-based finite-difference form of Eq. (S3), with $\epsilon = 1\times10^{-4}$; agreement with the exact eigenspectrum form is below 0.21 % across the USC sweep.

For the driven DCE protocol the state is not stationary, and the QFI is computed from the fidelity

$$F_Q(t) \approx 8\,\frac{1 - |\,\langle\Psi_\eta(t)|\Psi_{\eta+\delta}(t)\rangle\,|}{\delta^2}, \qquad \delta = 1\times10^{-3}, \tag{S4}$$

which becomes exact as $\delta \to 0$. All DCE QFI values quoted in the main text are this fidelity-based quantity. At the single-qubit DCE optimum the instantaneous variance form ($4\,\mathrm{Var}[\hat{H}_{\rm int}]$) gives 225.5, against the fidelity-based peak of 170 903; the two are different quantities and are not interchanged. Convergence with respect to time step and δ is reported in Section S2.

All values are normalized to the coherent-state baseline $F_Q^{\rm coh} = 1$. This baseline does not match the pump power, intracavity population, bandwidth, or readout cost of any driven comparison.

B. Energy budget

The exact static QFI obeys the reduced budget $F_Q \le 4(2n_{\rm virt}+1)$ with $n_{\rm virt} = \langle\hat{a}^\dagger\hat{a}\rangle_{\Psi_G}$. This bound holds at all 16 points of the regenerated USC sweep, with residuals $4(2n_{\rm virt}+1) - F_Q$ ranging from 2.50 ($\eta = 0.7$) to 10.39 ($\eta = 1.3$). The measured small-η asymptote $F_Q \approx 1 + 3\eta^2$ is consistent with $n_{\rm virt} \approx \eta^2/4$, so the metrological content is carried by the virtual population. Because the single qubit is a two-level system, this budget saturates; the measured curve peaks at $F_Q = 6.116$ at $\eta = 1.14$ and decreases beyond.

S2. NUMERICAL CONVERGENCE

A. Fock truncation

The Rabi QFI is converged to below 1×10^{-6} for $n_{\max} \geq 30$ at $\eta = 1.14$. The Dicke QFI for $N = 32$ is converged to below 0.01 % for $n_{\max} \geq 25$. The collective spin basis reduces the Hilbert space from 2^N to $N + 1$ and permits exact diagonalization at $N \leq 64$.

B. Time-step and finite-difference convergence of the DCE QFI

At the quoted single-qubit DCE operating point, the fidelity-based peak QFI varies by less than 0.15 % across time-step refinements $n_{\text{steps}} = 100$–2000 (quoted value at $n_{\text{steps}} = 500$). At fixed $n_{\text{steps}} = 500$, the peak varies by less than 0.40 % for δ between 1×10^{-4} and 5×10^{-4}, and by less than 0.12 % between $\delta = 1 \times 10^{-3}$ (quoted value) and $\delta = 1 \times 10^{-4}$. The quoted peak is therefore converged in both parameters.

S3. SINGLE-QUBIT DCE UNDER CAVITY LOSS

A. Loss sweep and retention

The single-qubit DCE protocol ($\eta = 0.150$, $g_d/g_0 = 0.119\,23$, $\omega_q/\omega_c = 1.027$, $\omega_d/\omega_c = 2.077$) is propagated under the Lindblad master equation with collapse operator $\sqrt{\kappa}\, a$, Fock truncation $N_{\text{cav}} = 20$, initial state $|0\rangle \otimes |\downarrow\rangle$, and fidelity-based QFI with $\delta = 1 \times 10^{-3}$. Table S1 reports the peak and late-window QFI versus κ.

The retention is not monotonic over the three lowest-loss rows (78 %, 79 %, 81 %). The first sampled row at or above 80 % retention is $\kappa/\omega_c = 1 \times 10^{-4}$ ($\kappa/2\pi = 0.6$ MHz); no continuous threshold is inferred. At the simulated physical operating point $\kappa/\omega_c = 4.1 \times 10^{-2}$ ($\kappa/2\pi \approx 246$ MHz; the cited CPW reference is approximately 248 MHz), retention is ~ 1 % of the pure-state peak. The degradation exceeds the simple envelope $\exp(-\kappa t)$ because the Lindblad dynamics modify the full state evolution, not just the peak amplitude.

TABLE S1. Loss-adjusted single-qubit DCE QFI. Reference pure-state peak: 170 903.

κ/ω_c	$\kappa/2\pi$ (MHz)	Peak QFI	Late-window avg	Percent of pure (peak)
0	0	170 903	132 800	100 %
1×10^{-5}	0.06	133 806	100 003	78 %
5×10^{-5}	0.3	134 693	100 977	79 %
1×10^{-4}	0.6	137 772	104 482	81 %
5×10^{-4}	3	119 926	96 606	70 %
1×10^{-3}	6	82 687	71 075	48 %
5×10^{-3}	30	11 653	6287	6.8 %
1×10^{-2}	60	5957	1919	3.5 %
2×10^{-2}	120	3376	853	2.0 %
4.1×10^{-2}	246	1640	406	0.96 %

B. Continuous κ sweep

A ten-point log-spaced sweep of $\kappa/\omega_c \in \{0, 1 \times 10^{-5}, 3 \times 10^{-5}, 1 \times 10^{-4}, 3 \times 10^{-4}, 1 \times 10^{-3}, 3 \times 10^{-3}, 1 \times 10^{-2}, 2 \times 10^{-2}, 4.1 \times 10^{-2}\}$ (500 output samples per point) plateaus at 0.78–0.81 retention for $\kappa/\omega_c \leq 3 \times 10^{-4}$, then falls to 0.484 at 1×10^{-3}, 0.117 at 3×10^{-3}, 0.035 at 1×10^{-2}, and 0.010 at the physical κ. Log-linear interpolation gives the crossovers $\kappa^*_{80} = 1.45 \times 10^{-4}$ and $\kappa^*_{50} = 9.38 \times 10^{-4}$. The 80 % crossover sits at the plateau edge and is mildly sensitive to the 1×10^{-4} bump; the 50 % crossover is robust to it.

C. Combined cavity, qubit-T_1, and temperature row

A single master-equation row combines, at the same operating point, the finite-temperature drive ($T = 20\,\mathrm{mK}$), physical cavity loss ($\kappa/\omega_c = 4.1 \times 10^{-2}$), and a conservative qubit-relaxation benchmark ($\gamma_1/\omega_c = 1 \times 10^{-4}$), with $N_{\mathrm{cav}} = 10$ over 20 drive periods. The pure-state peak is 30 418 and the joint peak is 1637, an end-to-end retention of 5.4 %. The joint loss-dominated peak matches the κ-sweep value at the same κ (1640) to within 0.2 %, confirming cavity decay as the dominant channel. The row is joint in (temperature $\times$ loss $\times$ T_1) at the two-level emitter truncation; a multilevel merge remains an open follow-up.

D. Combined cavity and qubit dissipation

The conservative transmon-like benchmark at $\omega_c/2\pi = 6\,\mathrm{GHz}$ ($T_1 = 30\,\mu\mathrm{s}$, $T_\phi = 5\,\mu\mathrm{s}$) gives dimensionless rates $\gamma_1/\omega_c = 8.84 \times 10^{-7}$ and $\gamma_\phi/\omega_c = 5.31 \times 10^{-6}$. Without cavity loss the qubit channels retain 78.56 % (T_1 alone), 79.08 % (T_ϕ alone), and 79.08 % (T_1 and T_ϕ together). At the physical cavity-loss point plus both qubit channels, retention is 0.9594 %, essentially identical to cavity loss alone. Cavity decay is the dominant channel over the tested window.

S4. COLLECTIVE DICKE+DCE UNDER CAVITY LOSS

A. Reduced-truncation loss matrix

The Dicke+DCE operating point ($N = 32$, $\eta = 1.15$, $g_d/g_0 = 0.15$, $\omega_q/\omega_c = 1.0$, $\omega_d/\omega_c = 2.0$) is propagated under Lindblad single-photon cavity loss at photon truncations $N_{\mathrm{cav}} = 6, 8, 10, 12$ and $\kappa/\omega_c \in \{0, 1 \times 10^{-4}, 1 \times 10^{-3}, 1 \times 10^{-2}, 4.1 \times 10^{-2}\}$, with `atol` = `rtol` = 1×10^{-10} and $n_{\mathrm{steps}} = 200$ over ten drive periods. Every stored state satisfies the trace, Hermiticity, and eigenvalue criteria (worst minimum eigenvalue -1.34×10^{-5} against the -1×10^{-4} floor).

TABLE S2. Loss-adjusted Dicke+DCE QFI at $N_{\mathrm{cav}} = 6$ (Hilbert dim 198). Reference pure-state peak: 2962.

κ/ω_c	$\kappa/2\pi$ (MHz)	Peak QFI	Late-window avg	Retained
0	0	2962	2594	:
1×10^{-4}	0.6	2927	2566	99 %
1×10^{-3}	6	2828	2490	95 %
1×10^{-2}	60	1818	1678	61 %
4.1×10^{-2}	246	474	321	16 %

B. Truncation dependence

Retention is truncation-dependent and not converged. The full trend at the physical κ is shown in Table S3.

TABLE S3. Truncation dependence of Dicke+DCE QFI retention at the physical cavity linewidth. Retention is relative to each truncation's own zero-loss peak.

N_{cav}	Hilbert dim	Peak QFI ($\kappa = 0$)	Peak QFI (physical κ)	Retention
6	198	2962	474	16 %
10	330	2363	290	12 %
12	396	4833	320	7 %
16	528	5786	255	4.4 %
20	660	9851	265	2.7 %
24	792	10 584	273	2.6 %

The non-monotonic pure-state peak (2962 → 2363 → 4833 → 5786 → 9851 → 10 584) reflects Fock-space confinement at the smallest truncation. Retention decreases monotonically. Of the six truncation comparisons against the larger truncation ($N_{\mathrm{cav}} = 16 \to 24$ and $20 \to 24$ at each of the three loss values), two meet the 5 % tolerance ($16 \to 24$ at $\kappa/\omega_c = 1 \times 10^{-2}$, $20 \to 24$ at the physical κ); the rest exceed it. The physical-point retention is therefore a truncation-dependent range of roughly 16 % ($N_{\mathrm{cav}} = 6$) down to 2.6 % ($N_{\mathrm{cav}} = 24$), not a converged number.

C. Boundary-peak convergence

Extending the pure-state Dicke+DCE time trace at $N_{\mathrm{cav}} = 6$ to periods 20–60 shows the peak QFI continues to grow monotonically and occurs at the last stored time point for every window (Table S4). The quoted boundary value 251 557× at $N_{\mathrm{cav}} = 28$ and $T_{\max} = 40$ is therefore an arbitrary cutoff, not a converged local maximum.

TABLE S4. Dicke+DCE boundary-peak convergence at $N_{\mathrm{cav}} = 6$. The peak occurs at the last stored time point for every window.

$T_{\max}$ (periods)	Peak QFI	Late-window avg
20	9932	8219
30	19 093	16 060
40	33 109	27 535
50	48 328	39 760
60	69 475	57 117

D. Photon-sector decomposition

Replaying the validated Dicke+DCE rows and projecting onto the $n \leq n_{\mathrm{cut}}$ sector shows the low-photon blocks carry the parameter-relevant coherence. At $N_{\mathrm{cav}} = 6$ ($\langle n \rangle = 2.9$–3.5) the $n \leq 2$ sector peak exceeds the full-space peak (2.65× at $\kappa = 0$, 1.28× at the physical κ) while retaining only 33 %–37 % trace weight; the overshoot decreases monotonically to 1.000 as $n_{\mathrm{cut}} \to N_{\mathrm{cav}}$. At $N_{\mathrm{cav}} = 12$ the sector curve converges monotonically with the cutoff (Table S5), and at the physical κ the $n_{\mathrm{cut}} = 10$ projection (84 % weight) is within 0.1 % of the full peak.

TABLE S5. Fraction of the full-space peak QFI retained by the $n \leq n_{\mathrm{cut}}$ photon sector at $N_{\mathrm{cav}} = 12$; trace weight in parentheses.

n_{cut}	4	6	8	10	12
$\kappa/\omega_c = 1 \times 10^{-2}$	0.791 (0.34)	0.860 (0.51)	0.897 (0.67)	0.946 (0.83)	1.000 (1.00)
$\kappa/\omega_c = 4.1 \times 10^{-2}$	0.753 (0.38)	0.925 (0.54)	0.988 (0.69)	1.001 (0.84)	1.000 (1.00)

Sector QFI is not additive across sectors, so the $N_{\mathrm{cav}} = 6$ overshoot is a projection effect. The curves are a truncation-sensitivity diagnostic, not a decomposition of the full QFI into independent contributions.

S5. CAVITY-CHANNEL HEOM CROSS-CHECK

The Markovian cavity-loss treatment underlying the retention values in Sections S3 and S4 is cross-checked against a cavity-mode HEOM calculation: a Fock-space mode ($d = 12$) coupled through the position quadrature $\hat{x} = \hat{a} + \hat{a}^\dagger$ to a structured Drude–Lorentz bath and benchmarked against $\sqrt{\kappa}\hat{a}$ Lindblad. The grid spans $\kappa/\omega_c \in \{1\times10^{-4}, 4.1\times10^{-2}\}$, Drude cutoffs $\gamma/\omega_c \in \{5, 10, 20, 40\}$, hierarchy depths $L = 2, 3, 4$, temperatures $T = 0.01, 0.05$, and windows $t_{\mathrm{final}} = 1, 2$ (96 rows).

At $\kappa/\omega_c = 1 \times 10^{-4}$ all 18 physical rows give mean fidelity errors of 9.2×10^{-5}–3.7×10^{-4}; the hierarchy shows no depth dependence, confirming that Markovian and non-Markovian dynamics are indistinguishable at this linewidth. At $\kappa/\omega_c = 4.1 \times 10^{-2}$ the 18 physical rows give mean fidelity errors of 4.8 %–24 % depending on the Drude cutoff and time window; the hierarchy converges at depth $L = 3$. The 60 remaining rows fail the positivity criterion (minimum eigenvalue down to -1.4), documenting a genuine numerical limitation of the HEOM hierarchy for strong cavity-bath coupling.

The Markovian treatment is therefore approximate at the physical linewidth: the retention values should be read as first-order Markovian estimates whose sensitivity to cavity-bath memory is a tested model-comparison range of approximately 4.8 %–24 %, not a propagated statistical uncertainty.

S6. MULTILEVEL-EMITTER LEAKAGE

The two-level approximation used in the main-text QFI computations is tested against a common oscillator-based emitter interface parameterized by declared transition-frequency ratios for three effective spectra: transmon-like, flux-qubit-like, and fluxonium-like. Each retains $d_e = 2, \ldots, 5$ emitter levels. Static rows use $\eta = 0.15$ and 1.0; driven rows use drive coefficients $g_d = 0.119\,23$ (reference) and 0.5 (strong) at $\kappa/\omega_c = 0$ and 4.1×10^{-2} over 40 drive periods. The adequacy criterion is declared before scanning: maximum leakage to levels $|2\rangle$ and above below 1 % over the window.

No driven $d_e \geq 3$ row meets the criterion at the reference drive. Window-maximum leakage ranges from 2.99 % (fluxonium-like, $d_e = 3$) to 81.5 % (fluxonium-like, $d_e = 5$, strong drive). Paired two-level versus multilevel comparisons show the multilevel peak QFI relative

change from the two-level baseline ranging from -0.873 to $+7.36$, with the largest positive deviations at strong drive where leakage is largest.

A staged drive-amplitude sweep over $g_d \in \{0, 0.02, 0.04, 0.06, 0.08, 0.10, 0.119\,23\}$ at $d_e = 3$ (42 rows, $\kappa/\omega_c = 0$ and 4.1×10^{-2}) identifies platform-specific safe operating windows at physical loss (Table S6). The QFI enhancement at those leakage-acceptable points is orders of magnitude below the reference value.

TABLE S6. Maximum drive coefficient g_d satisfying the 1 % leakage criterion at physical loss ($d_e = 3$, 40 drive periods) and the corresponding peak QFI for the coupling estimand.

Platform	Max g_d	Peak QFI
Fluxonium-like	0.06	85.5
Flux-qubit-like	0.04	77.5
Transmon-like	0.02	54.2

The trade-off between leakage and QFI enhancement is monotonic. The 1 % criterion is a model choice, and the models are declared effective spectra, not device simulations.

S7. FINITE-SIZE SCALING OF THE DICKE MODEL

The exact static Dicke ground-state QFI is evaluated at two cavity truncations per size bracket for $N = 4, \ldots, 64$ at $\omega_q/\omega_c = 1.0$ and 1.005 (52 rows). All static rows are truncation-converged (relative change below 5 %; typical values 10^{-11}–10^{-14}). The static QFI at $N = 32$, $\omega_q/\omega_c = 1.0$ is 228.6, reproducing the reference value.

Descriptive power-law fits over $N \in [6, 18]$ give $\alpha = 2.197 \pm 0.039$ ($\omega_q/\omega_c = 1.0$, $R^2 = 0.9984$) and $\alpha = 2.216 \pm 0.034$ ($\omega_q/\omega_c = 1.005$, $R^2 = 0.9988$), consistent with the historical $\alpha = 2.21$. Dropping the largest in-range point changes α by only 0.004. Beyond $N \approx 21$ the local growth exponent falls below unity (0.67 at $N = 24$, 0.34 at $N = 27$) before recovering to 0.85 at $N = 64$. Single-plateau saturated forms do not describe the data over $N \in [4, 64]$, so no saturation plateau is reached up to $N = 64$ within this grid. The fits are recorded with uncertainty intervals and boundary sensitivity; they are explicitly descriptive.

Driven Dicke+DCE rows at $N_{\mathrm{cav}} = 6$ show the lossless peak growing from 785 ($N = 4$) to 1852 ($N = 12$) and 2768 ($N = 24$) at a fixed pump proxy $g_d^2 = 0.0298$. Cavity loss reduces

the peak and moves it into the window (Table S7).

TABLE S7. Driven Dicke+DCE resource-accounting at $g_d^2 = 0.0298$ ($N_{\mathrm{cav}} = 6$). Emitted proxy is $\kappa \int n(t)\, dt$, not a measured output.

N	κ/ω_c	Peak F_Q	Peak period	Max $\bar{n}$	Emitted proxy
4	0	785.3	10.0	3.25	0
4	1×10^{-2}	469.3	10.0	3.17	0.663
4	4.1×10^{-2}	96.0	6.8	3.06	2.394
12	0	1851.6	10.0	3.19	0
12	1×10^{-2}	1072.6	10.0	2.90	0.696
12	4.1×10^{-2}	188.6	7.7	2.70	2.830
24	0	2768.1	10.0	2.94	0

The screen supports a descriptive superlinear finite-size trend over the declared interval, with uncertainty and boundary sensitivity reported; it establishes no thermodynamic exponent, no phase transition, and no restored global bound. The resource accounting is a partial pump/emitted-photon proxy comparison, not a jointly matched operational advantage.

Companion eta-sweep curves at $N = 4, 8, 16, 32$ (produced by the same physics core and anchored at $\eta = 1.15$ against the finite-size grid to below 3×10^{-7} relative deviation) show the sampled static-QFI maximum shifting with N (≈ 1.40 at $N = 4$, ≈ 1.15 at $N = 16$, ≈ 1.10 at $N = 32$). The tabulated $\eta = 1.15$ operating point is therefore not the sampled optimum for every N.

S8. ADDITIONAL MODEL-SPECIFIC SCREENS

The following screens bound the operational significance of the main-text QFI values but are not the central results. They are grouped here to keep the main text focused.

A. Thermal-competition screen

A finite-temperature Lindblad screen over $T \in [10, 200]$ mK and $\omega_c/2\pi \in [5, 8]$ GHz uses explicit emission and absorption channels $L_- = \sqrt{\kappa(n_{\mathrm{th}} + 1)}a$, $L_+ = \sqrt{\kappa n_{\mathrm{th}}}a^\dagger$ with

n_{th} evaluated at the declared cavity frequency, plus separate qubit-thermalization rows. The retention criterion is the first sampled temperature at which peak QFI relative to the same-configuration zero-temperature reference falls below 80 %.

For the reference driven row ($\eta = 0.15$, vacuum initial state, $\kappa/\omega_c = 4.1 \times 10^{-2}$, $\omega_c/2\pi = 6\,\mathrm{GHz}$), the zero-temperature peak QFI is 1642.1, retention is 0.985 at 50 mK and 0.775 at 100 mK, so the sampled crossover is at 100 mK. The cavity Bose–Einstein occupation is 5.6×10^{-7} at 20 mK and 5.95×10^{-2} at 100 mK. The screen is a sampled retention criterion under the declared bath and preparation; it is not a universal temperature threshold.

B. Homodyne output-field CFI screen

An idealized joint-system QFI does not by itself specify a laboratory readout. A bounded output-field screen uses the input–output convention $a_{\mathrm{out}}(t) = a_{\mathrm{in}}(t) + \sqrt{\kappa_{\mathrm{ext}}}a(t)$ with vacuum input and $\kappa = \kappa_{\mathrm{ext}} + \kappa_{\mathrm{int}}$; only the external fraction is treated as monitored output. The measured temporal mode is flat over 6 or 10 drive periods, with a Gaussian quadrature distribution. Among rows with a valid same-state comparison, F_C/F_Q ranges from 0 to 0.603; the largest static ratio is 0.0455. Historical proposals of 92.4 % static and 88.1 % driven are not reproduced. This is a finite-model Gaussian-readout result, not optimized matched-filter CFI or experimental QFI saturation.

C. Multiparameter QFI and representation consistency

Directional sensitivity is evaluated for the parameter vector $(\eta, \omega_q/\omega_c, g_d/g_0, \phi)$ in the driven protocol and $(\eta, \omega_q/\omega_c)$ for static ground states, using phase-aligned central finite differences at steps 1×10^{-2}–1×10^{-4}. All 18 rows are finite, positive semidefinite within tolerance, and converged. At the reference 10-period row and $N_{\mathrm{cav}} = 20$, the coupling diagonal is approximately 1.15×10^3 and the nuisance-projected effective coupling information is approximately 4.08×10^2. Driven condition numbers are of order 1×10^3.

A controlled similarity transformation $U = \exp[-i0.37a^\dagger a]$ at $N_{\mathrm{cav}} = 30$ transforms the Hamiltonian, initial state, control, and observables together; unitarity error is 1.6×10^{-15}, transformed-state infidelity below 4×10^{-16}, and the relative QFI-metric difference is 4.8×10^{-9}. This verifies the implemented finite-dimensional similarity transformation; it does

not establish agreement between arbitrary laboratory gauges with independently specified controls.

D. Interrogation-window crossover

At the fixed single-qubit DCE operating point, the fidelity-based QFI and the information-rate objective $R_Q(t) = F_Q(t)/t$ are evaluated over 40 drive periods for $\kappa/\omega_c \in \{0, 1\times10^{-4}, 1\times 10^{-3}, 5\times10^{-3}, 1\times10^{-2}, 2\times10^{-2}, 4.1\times10^{-2}\}$. At $\kappa/\omega_c \leq 1\times10^{-3}$ both metrics remain boundary-limited; at larger loss the R_Q maximum moves earlier (18.08, 14.00, 10.96, 6.96 periods at $\kappa/\omega_c = 0.005, 0.01, 0.02, 0.041$) with corresponding F_Q falling from 10 449 to 1242 (Table S8). The stored entropy derivative is a finite-difference mixedness diagnostic, not a thermodynamic entropy-production rate; the crossover is model-specific, not a universal or thermodynamically derived optimum.

TABLE S8. Information-rate maxima for the single-qubit DCE protocol.

κ/ω_c	$\kappa/2\pi$ (MHz)	t/T_d at R_Q max	F_Q at max	Window status
0	0	40.00	170 670	boundary
1×10^{-4}	0.6	40.00	161 087	boundary
1×10^{-3}	6	40.00	84 830	boundary
5×10^{-3}	30	18.08	10 449	interior
1×10^{-2}	60	14.00	5510	interior
2×10^{-2}	120	10.96	2961	interior
4.1×10^{-2}	246	6.96	1242	interior

E. Matched-photon and adaptive-protocol references

At matched intracavity photon budget $n = 9.01$ (the DCE peak), the pure-state DCE QFI is 4742× the coherent-state shot-noise reference $F_Q = 4n$ and 249× the squeezed-state reference $F_Q = 4n(1 + 2n)$. Photon-number matching does not match pump power, bandwidth, preparation, loss, readout, or observation time, so these ratios are not device-level advantages. Under cavity loss the DCE QFI collapses as quantified in Section S3.

A formal comparison with adaptive protocols is retained only as a reference at the DCE peak photon budget and the DCE time-average budget. The adaptive entries are uncertainty-bound or formal-reference expressions, whereas the DCE entries are QFI values, so no cross-estimand enhancement ratios are reported. A physically relevant comparison would require a common estimation task, prior, observation time, loss model, drive budget, and readout; that comparison is outside the present study.

F. Finite-window Floquet-stability screen

A deterministic screen at the driven single-qubit Rabi model ($\omega_q/\omega_c = 1.027$, $g(t) = g_0[1 + (g_d/g_0)\cos(\omega_d t + \phi)]$, cavity loss κ) evaluates coupling-estimand QFI from paired trajectories at η and $\eta+\delta\eta$, with $\delta\eta = 1\times10^{-3}$. Tier 1 uses $N_{\mathrm{cav}} = 12, 20$ and 40 periods over 108 parameter rows; Tier 2 validates five representative rows at 80 periods and $N_{\mathrm{cav}} = 20, 30$. All trajectories satisfy the trace, Hermiticity, and eigenvalue criteria.

Tier 1 classifies 79 rows as showing boundary growth and 29 as bounded within the screen; no row exceeds the 10 % truncation-difference criterion. Tier 2 resonant/low-loss rows continue to grow at the boundary, whereas two selected lossy rows remain bounded and agree between the two truncations to better than 10^{-6} in peak QFI. The persistence of boundary growth in low-loss resonant rows rejects the claim that two-level saturation is a general passive Floquet stabilizer. The screen is finite-window; no asymptotic stability or phase stability is established.

G. HEOM cross-checks (qubit bath)

Two HEOM comparisons bound the Lindblad working model for the qubit-bath channel.

Local short-time check. A Drude–Lorentz bath ($\omega_q = 1.0$, $\gamma_{\mathrm{cut}} = 20$, $\lambda = 0.02$, $\Gamma = 0.012\,535$, $T = 0.001$, hierarchy depth 3, 50 time points to $t_{\mathrm{final}} = 1.0$) gives a mean Uhlmann-fidelity error of 1.18 % (maximum 2.43 %), all sampled points below 5 %.

Extended check. A hierarchy-depth, temperature, coupling, cutoff, and window sweep (18 rows, 15 physical, 3 inconclusive) confirms hierarchy convergence at $L = 2$ (relative change 4.3×10^{-6} from $L = 2$ to 5). Baseline agreement is 5.3×10^{-4} mean, 6.6×10^{-4} maximum. Two driven DCE-ratio rows give QFI relative differences of 1.05 % and 1.09 %

at $\eta_d = 0.15$ and 0.10. The $T = 0.35$ row shows a larger mean error (7.7 %), retained as a temperature-scope caveat.

Neither HEOM check validates the driven high-QFI traces; both support the working Lindblad model for their declared scope only.

S9. EXPERIMENTAL FEASIBILITY

Table S9 compares the model target with the demonstrated state of the art at the key parameters.

TABLE S9. Experimental feasibility assessment.

Requirement	**Model target**	**CPW (current USC)**	**3D cavity (candidate)**
Number of qubits N	32	4 [3]	1 [4]
Coupling η	1.15	1.3 [5]	0.1 to 0.3 [4]
Coupling $g/2\pi$	6.9 GHz	6 GHz [5]	0.3 GHz [4]
DCE drive $g_d/2\pi$	1.0 GHz	,	,
Drive frequency $\omega_d/2\pi$	12 GHz	Standard	Standard
Temperature	< 200 mK	10 mK	10 mK
Cavity $\kappa/2\pi$	< 1 MHz	248 MHz	0.1 MHz
Coherence T_1	$\gg 83$ ns	10 µs to 100 µs	10 µs to 100 µs
Floquet bound-state rescue demonstrable at $\kappa/\omega \sim 4 \times 10^{-2}$		emitter-local only [6]	unshown for cavity leakage

No demonstrated platform simultaneously satisfies (i) the $\eta = 1.15$ coupling needed for the superradiant collective enhancement, (ii) the sampled single-qubit reference point $\kappa/2\pi = 0.6$ MHz at which the loss sweep first reaches 80.6 % retention, and (iii) the uniform collective coupling of $N = 32$ qubits. The cited CPW example provides strong coupling and a partial collective architecture at $\kappa/2\pi \approx 248$ MHz; the cited 3D-cavity example provides the low-linewidth regime at the single-qubit level.

Platform candidates. Flux qubits are the original USC platform with tunable coupling, best for $N = 1$ to 4 [5]. Transmons are less anharmonic but more coherent, reaching USC with flux-tunable couplers, best for $N > 8$ due to fabrication uniformity [3]. Fluxonium offers high anharmonicity and long coherence, promising for USC with $N = 1$ to 2 [7]. The

three platform schematics are shown in Figs. S1 to S3.

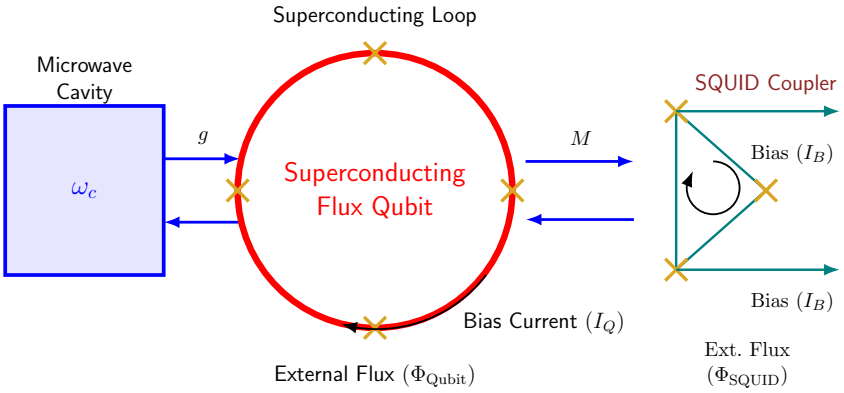


FIG. S1. Flux qubit platform. A superconducting loop with Josephson junctions forms a flux qubit (red) coupled to a microwave cavity (blue); external flux Φ_{ext} tunes the qubit frequency.

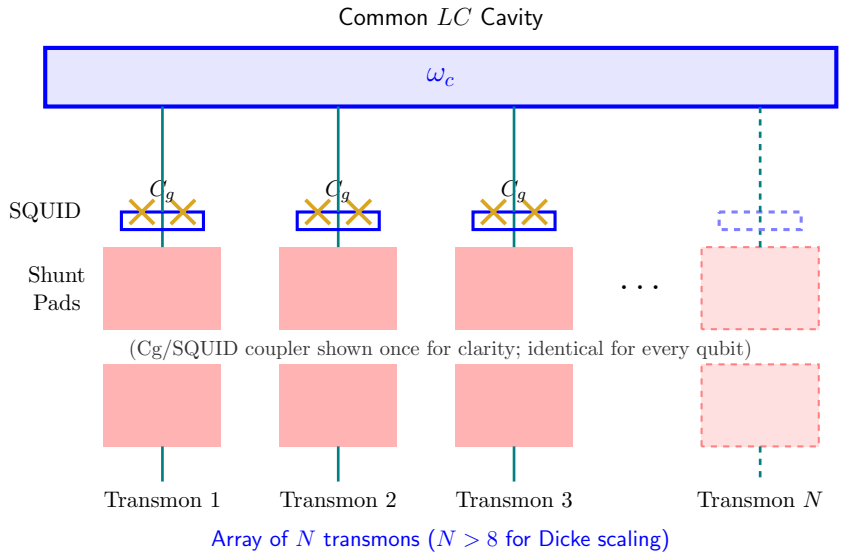


FIG. S2. Transmon platform. An array of transmon qubits (red) capacitively coupled to a common LC cavity (blue) via tunable couplers C_g.

Staged program. The experimental roadmap has three stages. Stage 1 ($N = 1$, single qubit, flux-based CPW or 3D cavity) benchmarks the cavity-loss-augmented QFI against the first sampled single-qubit reference row (80.6 % at $\kappa/2\pi = 0.6\,\text{MHz}$); a 3D cavity at $\omega_c/2\pi = 6\,\text{GHz}$ is a candidate platform. Stage 2 ($N = 4$ to 8) tests the collective scaling exponent in the loss-controlled regime at the quoted $\eta = 1.15$, without engaging the DCE

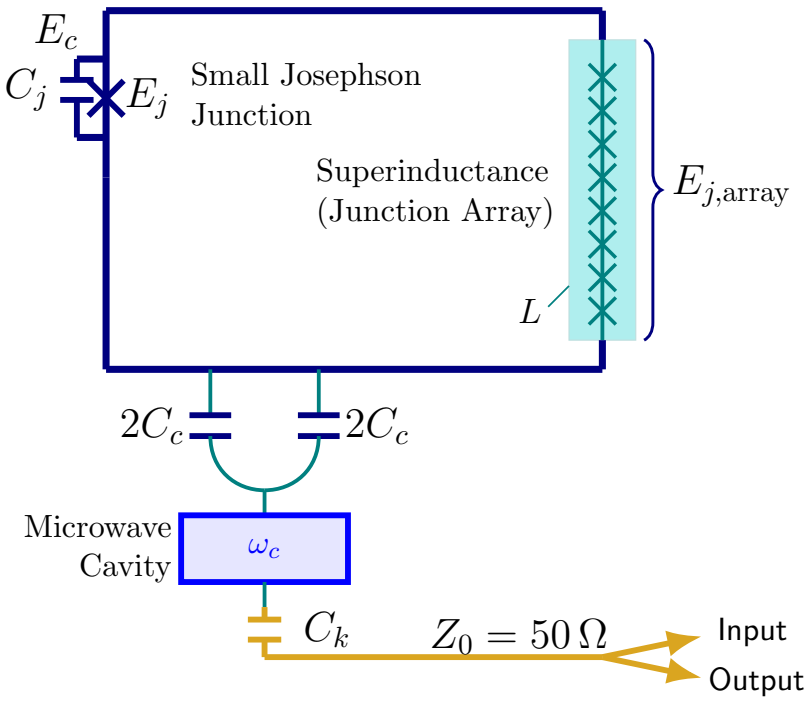


FIG. S3. Fluxonium platform. High anharmonicity and long coherence enable operation at $\eta \sim 1.3$ with $T_2 > 10\,\mu\text{s}$.

modulation. Stage 3 ($N = 16$ to 32) simultaneously demonstrates collective dressing, DCE modulation, and the loss barrier; a 3D-cavity-based array is a candidate platform for the required low linewidth, but whether it can also provide the required collective coupling uniformity remains open. None of the three stages has been demonstrated end-to-end.

A. Parameter-dependence scan

S10. SCOPE AND LIMITATIONS

The results in this Supplemental Material are confined to the stated finite-dimensional models, drive conventions, and dissipation channels. They do not establish:

- loss immunity of the virtual-dressing state (the static result concerns parameter sensitivity of an undriven dressed state, not an extracted output);
- a resource-normalized quantum advantage (photon-budget matching alone is insuffi-

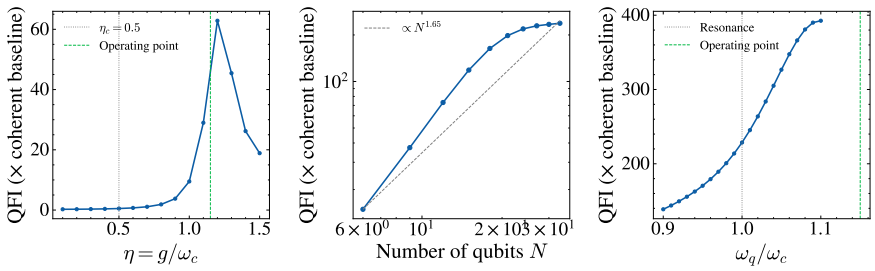


FIG. S4. Parameter sensitivity. (a) QFI vs η at $N = 10$, $n_{\max} = 15$ (exploratory): the finite-size superradiant regime above $\eta_c = 0.5$ shows a sharp onset, with $\sim 7\,\%$ variation for $\eta > 1.1$. (b) QFI vs N at $\eta = 1.15$, $\omega_q/\omega_c = 1.005$: the super-linear $\propto N^{2.22}$ fit on $N \leq 18$ gives $N = 16$ a factor $\sim 134\times$. (c) QFI vs detuning at $N = 32$, $n_{\max} = 35$: maximum at $\omega_q/\omega_c = 1.10$. Only panel (a) is exploratory; the retention claim refers to the detuning panel only.

cient);

- an experimentally demonstrated QFI enhancement (the cavity HEOM cross-check shows a 4.8 %–24 % model-comparison range at the physical linewidth);

- two-level-emitter validity at the drive amplitudes of the main-text DCE QFI (all driven $d_e \geq 3$ rows fail the 1 % leakage criterion at the reference drive; safe operating windows exist only at reduced drive with correspondingly reduced enhancement);

- a truncation-converged Dicke+DCE retention (the reduced-truncation range is 2.6 %–16 % across $N_{\text{cav}} = 6$–24);

- a thermodynamic or universal interrogation optimum (the crossover is model-specific);

- a universal temperature threshold (the sampled crossover is tied to the explicit cavity frequency and bath model);

- a general passive Floquet stabilizer (the finite-window screen retains boundary growth in low-loss resonant rows);

- gauge-independent device predictions from the effective time-dependent Hamiltonian (the control convention must be specified across gauges).

DATA AVAILABILITY

Manuscript sources, deterministic analysis scripts, and the environment specification are publicly archived at https://github.com/NanaEngo/Floquet-virtual-state-metrology-dce.

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